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**MODEL :KAL-1600-SG  
ANODIZING LINE**

# **API USER MANUAL**

**Project No. : PD-03726**

**Date : 29-SEP-2026**

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# KAN-PACIFIC FX5U PLC Polling Data API

## User Manual

Version 1.0 | How to start, authenticate, read, and write PLC tags

### 1. Overview

This API provides HTTP access to Mitsubishi FX5U PLC tag data. The poller (plc\_polling.py) continuously reads tags defined in TagsMap.xlsx and stores them in plc\_polling\_data.db. The API (plc\_api.py) serves the latest values and can write to writable tags on the PLC.

#### Data flow:

TagsMap.xlsx → plc\_polling.py → plc\_polling\_data.db → plc\_api.py → HTTP clients  
📡 write path 📡 (direct PLC connection for W tags)

### 2. Prerequisites

- Python virtual environment with project dependencies installed (plc\_env).
- TagsMap.xlsx present in the project root.
- Network access to the FX5U PLC (IP/port in plc\_config.py).
- At least one successful poller run so plc\_polling\_data.db exists.

### 3. Getting Started

#### 3.1 Activate the environment

```
.\plc_env\Scripts\activate  
pip install -r requirements.txt
```

#### 3.2 Generate an API key

Protected endpoints require the X-API-Key header. Generate a development key:

```
python generate_api_key.py
```

This creates/updates dev\_api\_key.txt in the project root (gitignored). Do not commit this file.

#### 3.3 Start the poller

In one terminal, start continuous polling:

```
python plc_polling.py
```

On startup the poller resets any stale PLC connection lock left by an abnormal shutdown, then polls tags and updates the database.

## 3.4 Start the API server

In another terminal:

```
python plc_api.py
```

Default base URL: <http://127.0.0.1:8000>

Interactive Swagger docs: <http://127.0.0.1:8000/docs>

## 4. Authentication

Send the API key in every protected request:

X-API-Key: <your-key>

Key resolution order:

- Environment variable PLC\_API\_KEY
- File dev\_api\_key.txt in the project root

| HTTP Status | Meaning                                          |
|-------------|--------------------------------------------------|
| 401         | Missing or invalid API key                       |
| 503         | API key not configured — run generate_api_key.py |

## 5. Configuration

| Variable     | Default              | Description             |
|--------------|----------------------|-------------------------|
| PLC_API_KEY  | from dev_api_key.txt | API key value           |
| PLC_DB_PATH  | plc_polling_data.db  | Path to SQLite database |
| PLC_API_HOST | 127.0.0.1            | Bind address            |
| PLC_API_PORT | 8000                 | Listen port             |

PowerShell example to listen on all interfaces:

```
$env:PLC_API_HOST = "0.0.0.0"
$env:PLC_API_PORT = "8080"
python plc_api.py
```

## 6. API Endpoints

### 6.1 GET /api/v1/tags/list

List all tags with metadata used by clients for display and write validation.

Auth: required

Response fields per tag:

| Field    | Description                 |
|----------|-----------------------------|
| tag_name | Tag key, e.g. Tank01_TempSV |



|                |                                                              |
|----------------|--------------------------------------------------------------|
| device_address | PLC address, e.g. D4000 or D1002.8                           |
| data_type      | Bool, Word, SignedWord, or Float                             |
| read_write     | R = read-only, W = writable                                  |
| multiplier     | Engineering scale factor (read: raw×mult, write: value÷mult) |
| description    | Human-readable text; falls back to tag_name if empty         |
| low_limit      | Present only when set in TagsMap.xlsx                        |
| high_limit     | Present only when set in TagsMap.xlsx (HighLimit/UpLimit)    |

Example (PowerShell):

```
$key = (Get-Content dev_api_key.txt).Trim()
Invoke-RestMethod "http://127.0.0.1:8000/api/v1/tags/list" -Headers @{"X-API-Key"=$key}
```

Example response (abbreviated):

```
{
  "tags": [
    {
      "tag_name": "Tank01_TempSV",
      "device_address": "D4000",
      "data_type": "Word",
      "read_write": "W",
      "multiplier": 0.1,
      "description": "Tank01 Temperature Set Value",
      "low_limit": 0.0,
      "high_limit": 999.9
    }
  ]
}
```

## 6.2 GET /api/v1/tags/read

Return the most recent poll cycle from the database, including engineering values for every tag. Description and limits are not included here.

Auth: required

**Common response fields:**

| Field       | Description                                          |
|-------------|------------------------------------------------------|
| id          | Database poll row id                                 |
| timestamp   | ISO timestamp of the poll                            |
| poll_count  | Cyclic poll slot (1..MAX_POLL_CYCLES)                |
| tags_ok     | Number of successful tag reads                       |
| tags_failed | Number of failed tag reads                           |
| <tag_name>  | Engineering value for each tag (null if read failed) |

Example:

```
Invoke-RestMethod "http://127.0.0.1:8000/api/v1/tags/read" -Headers @{"X-API-Key"=$key}
```

## 6.3 POST /api/v1/tags/write

Write one engineering value to a writable tag (ReadWrite=W). The API acquires the shared PLC lock, connects to the PLC, writes, then releases the lock.

Auth: required

**Request body:**

```
{
  "tag_name": "Tank01_TempSV",
  "value": 25.0
}
```

**Value rules:**

- Bool: 0 or 1 only
- Word / SignedWord / Float: engineering units (after Multiplier scaling to PLC raw)
- If LowLimit/HighLimit exist in TagsMap, value must be within range

**Example:**

```
$body = @{ tag_name = "Tank01_TempSV"; value = 25.0 } | ConvertTo-Json
Invoke-RestMethod "http://127.0.0.1:8000/api/v1/tags/write" `
  -Method Post -Headers @{ "X-API-Key"=$key; "Content-Type"="application/json" } `
  -Body $body
```

Success response fields include tag\_name, device\_address, data\_type, read\_write, multiplier, value (engineering), and written=true.

## 6.4 POST /api/v1/tags/write/batch

Write multiple writable tags in one PLC session (1–100 items). Duplicate tag names in the same batch are rejected.

**Request body:**

```
{
  "writes": [
    { "tag_name": "Tank01_TempSV", "value": 25.0 },
    { "tag_name": "Tank01_HeaterOn", "value": 1 }
  ]
}
```

**Example:**

```
$body = @(
  writes = @(
    @{ tag_name = "Tank01_TempSV"; value = 25.0 },
    @{ tag_name = "Tank01_HeaterOn"; value = 1 }
  )
) | ConvertTo-Json -Depth 4
Invoke-RestMethod "http://127.0.0.1:8000/api/v1/tags/write/batch" `
  -Method Post -Headers @{ "X-API-Key"=$key; "Content-Type"="application/json" } `
  -Body $body
```

## 7. TagsMap Concepts Affecting the API

### 7.1 Multiplier

Engineering value = PLC raw × Multiplier on read. On write, PLC raw = round(engineering ÷ Multiplier) for Word and SignedWord tags. Example: Multiplier 0.1, write 25.0 → PLC register 250; read 250 → API 25.0. SignedWord also accepts negatives (write -5.0 → PLC register -50).

### 7.2 Data types

| Data Type  | Behavior                                             |
|------------|------------------------------------------------------|
| Bool       | Bit devices, or bit-of-word addresses like D1002.8   |
| Word       | Unsigned 16-bit register (0..65535 after scaling)    |
| SignedWord | Signed 16-bit register (-32768..32767 after scaling) |
| Float      | IEEE-754 REAL over 2 consecutive D registers         |

### 7.3 Limits

LowLimit and HighLimit in TagsMap.xlsx are enforced on write using engineering units. Out-of-range values return HTTP 400. Blank limits mean no bound for that side.

### 7.4 Descriptions

Returned by /tags/list. Tank tags are prefixed with the tank column name (e.g. “Tank01 Temperature Set Value”). Empty Description falls back to tag\_name.

## 8. Tag Catalog (from TagsMap.xlsx)

Complete list of tags currently loaded from TagsMap.xlsx (362 tags: 141 writable, 221 read-only).

Regenerate this manual after TagsMap changes to keep the list current.

| Tag Name | Address | Type | R/W | Mult | Description                         | Low | High |
|----------|---------|------|-----|------|-------------------------------------|-----|------|
| Alarm01  | M100    | Bool | R   | 1.0  | The power Supply phases are swapped |     |      |
| Alarm02  | M101    | Bool | R   | 1.0  | Emergency Stop                      |     |      |
| Alarm03  | M102    | Bool | R   | 1.0  | HMI Emergency Stop                  |     |      |
| Alarm04  | M103    | Bool | R   | 1.0  | 12# Emergency Stop                  |     |      |
| Alarm05  | M104    | Bool | R   | 1.0  | 2# Emergency Stop                   |     |      |
| Alarm06  | M105    | Bool | R   | 1.0  | 9# Emergency Stop                   |     |      |
| Alarm07  | M106    | Bool | R   | 1.0  | spare                               |     |      |
| Alarm08  | M107    | Bool | R   | 1.0  | spare                               |     |      |
| Alarm09  | M108    | Bool | R   | 1.0  | spare                               |     |      |
| Alarm10  | M109    | Bool | R   | 1.0  | spare                               |     |      |
| Alarm11  | M110    | Bool | R   | 1.0  | spare                               |     |      |
| Alarm12  | M111    | Bool | R   | 1.0  | spare                               |     |      |
| Alarm13  | M112    | Bool | R   | 1.0  | spare                               |     |      |

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|         |      |      |   |     |                          |  |  |
|---------|------|------|---|-----|--------------------------|--|--|
| Alarm14 | M113 | Bool | R | 1.0 | spare                    |  |  |
| Alarm15 | M114 | Bool | R | 1.0 | spare                    |  |  |
| Alarm16 | M115 | Bool | R | 1.0 | spare                    |  |  |
| Alarm17 | M116 | Bool | R | 1.0 | spare                    |  |  |
| Alarm18 | M117 | Bool | R | 1.0 | spare                    |  |  |
| Alarm19 | M118 | Bool | R | 1.0 | spare                    |  |  |
| Alarm20 | M119 | Bool | R | 1.0 | spare                    |  |  |
| Alarm21 | M120 | Bool | R | 1.0 | 1# Filter Pump Overload  |  |  |
| Alarm22 | M121 | Bool | R | 1.0 | 3# Filter Pump Overload  |  |  |
| Alarm23 | M122 | Bool | R | 1.0 | 5# Filter Pump Overload  |  |  |
| Alarm24 | M123 | Bool | R | 1.0 | 5# Swing Pump Overload   |  |  |
| Alarm25 | M124 | Bool | R | 1.0 | 7# Filter Pump Overload  |  |  |
| Alarm26 | M125 | Bool | R | 1.0 | 10# Filter Pump Overload |  |  |
| Alarm27 | M126 | Bool | R | 1.0 | 11# Filter Pump Overload |  |  |
| Alarm28 | M127 | Bool | R | 1.0 | 14# Filter Pump Overload |  |  |
| Alarm29 | M128 | Bool | R | 1.0 | 16# Filter Pump Overload |  |  |
| Alarm30 | M129 | Bool | R | 1.0 | 1# Heater                |  |  |
| Alarm31 | M130 | Bool | R | 1.0 | 3# Heater                |  |  |
| Alarm32 | M131 | Bool | R | 1.0 | 5# Heater                |  |  |
| Alarm33 | M132 | Bool | R | 1.0 | 7# Heater                |  |  |
| Alarm34 | M133 | Bool | R | 1.0 | 10# Heater               |  |  |
| Alarm35 | M134 | Bool | R | 1.0 | 13# Heater               |  |  |
| Alarm36 | M135 | Bool | R | 1.0 | 14# Heater               |  |  |
| Alarm37 | M136 | Bool | R | 1.0 | 16# Heater               |  |  |
| Alarm38 | M137 | Bool | R | 1.0 | Front Water Leak         |  |  |
| Alarm39 | M138 | Bool | R | 1.0 | Back Water Leak          |  |  |
| Alarm40 | M139 | Bool | R | 1.0 | spare                    |  |  |
| Alarm41 | M140 | Bool | R | 1.0 | spare                    |  |  |
| Alarm42 | M141 | Bool | R | 1.0 | spare                    |  |  |
| Alarm43 | M142 | Bool | R | 1.0 | spare                    |  |  |
| Alarm44 | M143 | Bool | R | 1.0 | spare                    |  |  |
| Alarm45 | M144 | Bool | R | 1.0 | spare                    |  |  |
| Alarm46 | M145 | Bool | R | 1.0 | spare                    |  |  |
| Alarm47 | M146 | Bool | R | 1.0 | spare                    |  |  |
| Alarm48 | M147 | Bool | R | 1.0 | spare                    |  |  |
| Alarm49 | M148 | Bool | R | 1.0 | spare                    |  |  |
| Alarm50 | M149 | Bool | R | 1.0 | spare                    |  |  |

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|                 |      |      |   |     |                            |  |  |
|-----------------|------|------|---|-----|----------------------------|--|--|
| Alarm51         | M150 | Bool | R | 1.0 | 1# tank level low          |  |  |
| Alarm52         | M151 | Bool | R | 1.0 | 1# buffer tank level high  |  |  |
| Alarm53         | M152 | Bool | R | 1.0 | 1# Buffer tank level low   |  |  |
| Alarm54         | M153 | Bool | R | 1.0 | 3# tank level low          |  |  |
| Alarm55         | M154 | Bool | R | 1.0 | 3# buffer tank level high  |  |  |
| Alarm56         | M155 | Bool | R | 1.0 | 3# Buffer tank level low   |  |  |
| Alarm57         | M156 | Bool | R | 1.0 | 5# tank level low          |  |  |
| Alarm58         | M157 | Bool | R | 1.0 | 5# buffer tank level high  |  |  |
| Alarm59         | M158 | Bool | R | 1.0 | 5# Buffer tank level low   |  |  |
| Alarm60         | M159 | Bool | R | 1.0 | 7# tank level low          |  |  |
| Alarm61         | M160 | Bool | R | 1.0 | 7# buffer tank level high  |  |  |
| Alarm62         | M161 | Bool | R | 1.0 | 7# Buffer tank level low   |  |  |
| Alarm63         | M162 | Bool | R | 1.0 | 10# tank level low         |  |  |
| Alarm64         | M163 | Bool | R | 1.0 | 10# buffer tank level high |  |  |
| Alarm65         | M164 | Bool | R | 1.0 | 10# Buffer tank level low  |  |  |
| Alarm66         | M165 | Bool | R | 1.0 | 11# tank level low         |  |  |
| Alarm67         | M166 | Bool | R | 1.0 | 13# tank level high        |  |  |
| Alarm68         | M167 | Bool | R | 1.0 | 13# tank level low         |  |  |
| Alarm69         | M168 | Bool | R | 1.0 | 14# tank level low         |  |  |
| Alarm70         | M169 | Bool | R | 1.0 | 14# buffer tank level high |  |  |
| Alarm71         | M170 | Bool | R | 1.0 | 14# Buffer tank level low  |  |  |
| Alarm72         | M171 | Bool | R | 1.0 | 16# tank level low         |  |  |
| Alarm73         | M172 | Bool | R | 1.0 | 16# buffer tank level high |  |  |
| Alarm74         | M173 | Bool | R | 1.0 | 16# Buffer tank level low  |  |  |
| Alarm75         | M174 | Bool | R | 1.0 | spare                      |  |  |
| Alarm76         | M175 | Bool | R | 1.0 | spare                      |  |  |
| Alarm77         | M176 | Bool | R | 1.0 | spare                      |  |  |
| Alarm78         | M177 | Bool | R | 1.0 | spare                      |  |  |
| Alarm79         | M178 | Bool | R | 1.0 | spare                      |  |  |
| Alarm80         | M179 | Bool | R | 1.0 | spare                      |  |  |
| Local           | L3   | Bool | R | 1.0 | Local (Manual Mode)        |  |  |
| MainInlet       | L190 | Bool | W | 1.0 | Main Inlet Always On       |  |  |
| MainInletStatus | Y5   | Bool | R | 1.0 | Main Inlet Status          |  |  |

|                        |         |            |   |     |                                                    |        |        |
|------------------------|---------|------------|---|-----|----------------------------------------------------|--------|--------|
| Remote                 | L4      | Bool       | R | 1.0 | Remote (API Mode)                                  |        |        |
| Tank01A_USSV           | D1044   | Word       | W | 0.1 | Tank01A Ultrasonic Generator Current Set Value (A) | 0.0    | 8.0    |
| Tank01B_USSV           | D1048   | Word       | W | 0.1 | Tank01B Ultrasonic Generator Current Set Value (A) | 0.0    | 8.0    |
| Tank01_DoorCloseRemote | M301    | Bool       | W | 1.0 | Tank01 Door Remote Close                           |        |        |
| Tank01_DoorClosed      | D1000.5 | Bool       | R | 1.0 | Tank01 Door Actually Closed                        |        |        |
| Tank01_DoorOpenRemote  | M300    | Bool       | W | 1.0 | Tank01 Door Remote Open                            |        |        |
| Tank01_DoorOpened      | D1000.4 | Bool       | R | 1.0 | Tank01 Door Actually Opened                        |        |        |
| Tank01_FillWaterAuto   | L200    | Bool       | W | 1.0 | Tank01 Fill Water Auto                             |        |        |
| Tank01_FillWaterOn     | L100    | Bool       | W | 1.0 | Tank01 Fill Water always On                        |        |        |
| Tank01_FillWaterStatus | D1002.8 | Bool       | R | 1.0 | Tank01 Fill Water Valve Current Status             |        |        |
| Tank01_FilterRemote    | M320    | Bool       | W | 1.0 | Tank01 Filter Remote On                            |        |        |
| Tank01_FilterStatus    | D1002.1 | Bool       | R | 1.0 | Tank01 Filter Pump Status                          |        |        |
| Tank01_HaveBasket      | M601    | Bool       | R | 1.0 | Tank01 The Tank has Basket                         |        |        |
| Tank01_HeaterRemote    | M321    | Bool       | W | 1.0 | Tank01 Heater Remote On                            |        |        |
| Tank01_HeaterStatus    | D1002.2 | Bool       | R | 1.0 | Tank01 Heater Status                               |        |        |
| Tank01_RemoteStart     | M323    | Bool       | W | 1.0 | Tank01 Remote Start                                |        |        |
| Tank01_TempAdj         | D4010   | SignedWord | W | 0.1 | Tank01 Temperature +- Adjust (Degree Celsius)      | -999.0 | 999.9  |
| Tank01_TempPV          | D4000   | Word       | R | 0.1 | Tank01 Temperature Present Value (Degree Celsius)  |        |        |
| Tank01_TempSV          | D4020   | Word       | W | 0.1 | Tank01 Temperature Set Value (Degree Celsius)      | 0.0    | 999.9  |
| Tank01_TimePV          | CN101   | Word       | R | 1.0 | Tank01 Time Present Value (s)                      |        |        |
| Tank01_TimeSV          | D4051   | Word       | W | 1.0 | Tank01 Time Set Value (s)                          | 0.0    | 9999.0 |

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|                           |         |       |   |     |                                        |     |        |
|---------------------------|---------|-------|---|-----|----------------------------------------|-----|--------|
| Tank01_TimeUp             | CS101   | Bool  | R | 1.0 | Tank01 Time Up                         |     |        |
| Tank01_USRemote           | M322    | Bool  | W | 1.0 | Tank01 Ultrasonic Remote On            |     |        |
| Tank01_USStatus           | Y4      | Bool  | R | 1.0 | Tank01 Ultrasonic Status               |     |        |
| Tank01_WaterFlow          | D1004   | Float | R | 1.0 | Tank01 Water Flow Meter Value (LPM)    |     |        |
| Tank02_AirAgitationAuto   | L202    | Bool  | W | 1.0 | Tank02 Air Agitation Auto              |     |        |
| Tank02_AirAgitationOn     | L102    | Bool  | W | 1.0 | Tank02 Air Agitation Always On         |     |        |
| Tank02_AirAgitationStatus | D1002.8 | Bool  | R | 1.0 | Tank02 Air Agitation Status            |     |        |
| Tank02_FillWaterAuto      | L201    | Bool  | W | 1.0 | Tank02 Fill Water Auto                 |     |        |
| Tank02_FillWaterOn        | L101    | Bool  | W | 1.0 | Tank02 Fill Water always On            |     |        |
| Tank02_FillWaterStatus    | D1002.C | Bool  | R | 1.0 | Tank02 Fill Water Valve Current Status |     |        |
| Tank02_HaveBasket         | M602    | Bool  | R | 1.0 | Tank02 The Tank has Basket             |     |        |
| Tank02_RemoteStart        | M324    | Bool  | W | 1.0 | Tank02 Remote Start                    |     |        |
| Tank02_TimePV             | CN102   | Word  | R | 1.0 | Tank02 Time Present Value (s)          |     |        |
| Tank02_TimeSV             | D4052   | Word  | W | 1.0 | Tank02 Time Set Value (s)              | 0.0 | 9999.0 |
| Tank02_TimeUp             | CS102   | Bool  | R | 1.0 | Tank02 Time Up                         |     |        |
| Tank02_WaterFlow          | D1006   | Word  | R | 0.1 | Tank02 Water Flow Meter Value (LPM)    |     |        |
| Tank03_DoorCloseRemote    | M303    | Bool  | W | 1.0 | Tank03 Door Remote Close               |     |        |
| Tank03_DoorClosed         | D1010.5 | Bool  | R | 1.0 | Tank03 Door Actually Closed            |     |        |
| Tank03_DoorOpenRemote     | M302    | Bool  | W | 1.0 | Tank03 Door Remote Open                |     |        |
| Tank03_DoorOpened         | D1010.4 | Bool  | R | 1.0 | Tank03 Door Actually Opened            |     |        |
| Tank03_FillWaterAuto      | L220    | Bool  | W | 1.0 | Tank03 Fill Water Auto                 |     |        |
| Tank03_FillWaterOn        | L110    | Bool  | W | 1.0 | Tank03 Fill Water always On            |     |        |
| Tank03_FillWaterStatus    | D1012.8 | Bool  | R | 1.0 | Tank03 Fill Water Valve Current Status |     |        |
| Tank03_FilterRemote       | M332    | Bool  | W | 1.0 | Tank03 Filter Remote On                |     |        |

|                           |         |            |   |     |                                                   |        |        |
|---------------------------|---------|------------|---|-----|---------------------------------------------------|--------|--------|
| Tank03_FilterStatus       | D1012.1 | Bool       | R | 1.0 | Tank03 Filter Pump Status                         |        |        |
| Tank03_HaveBasket         | M603    | Bool       | R | 1.0 | Tank03 The Tank has Basket                        |        |        |
| Tank03_HeaterRemote       | M333    | Bool       | W | 1.0 | Tank03 Heater Remote On                           |        |        |
| Tank03_HeaterStatus       | D1012.2 | Bool       | R | 1.0 | Tank03 Heater Status                              |        |        |
| Tank03_RemoteStart        | M330    | Bool       | W | 1.0 | Tank03 Remote Start                               |        |        |
| Tank03_SwingRemote        | M334    | Bool       | W | 1.0 | Tank03 Swing Remote On                            |        |        |
| Tank03_SwingSV            | D1008   | Word       | W | 1.0 | Tank03 Swing Speed Set Value (CPM)                | 5.0    | 120.0  |
| Tank03_SwingStatus        | D1012.4 | Bool       | R | 1.0 | Tank03 Swing Status                               |        |        |
| Tank03_TempAdj            | D4011   | SignedWord | W | 0.1 | Tank03 Temperature +- Adjust (Degree Celsius)     | -999.0 | 999.9  |
| Tank03_TempPV             | D4001   | Word       | R | 0.1 | Tank03 Temperature Present Value (Degree Celsius) |        |        |
| Tank03_TempSV             | D4021   | Word       | W | 0.1 | Tank03 Temperature Set Value (Degree Celsius)     | 0.0    | 999.9  |
| Tank03_TimePV             | CN103   | Word       | R | 1.0 | Tank03 Time Present Value (s)                     |        |        |
| Tank03_TimeSV             | D4053   | Word       | W | 1.0 | Tank03 Time Set Value (s)                         | 0.0    | 9999.0 |
| Tank03_TimeUp             | CS103   | Bool       | R | 1.0 | Tank03 Time Up                                    |        |        |
| Tank03_WaterFlow          | D1014   | Float      | R | 1.0 | Tank03 Water Flow Meter Value (LPM)               |        |        |
| Tank04_AirAgitationAuto   | L212    | Bool       | W | 1.0 | Tank04 Air Agitation Auto                         |        |        |
| Tank04_AirAgitationOn     | L112    | Bool       | W | 1.0 | Tank04 Air Agitation Always On                    |        |        |
| Tank04_AirAgitationStatus | D1012.8 | Bool       | R | 1.0 | Tank04 Air Agitation Status                       |        |        |
| Tank04_FillWaterAuto      | L211    | Bool       | W | 1.0 | Tank04 Fill Water Auto                            |        |        |
| Tank04_FillWaterOn        | L111    | Bool       | W | 1.0 | Tank04 Fill Water always On                       |        |        |
| Tank04_FillWaterStatus    | D1012.C | Bool       | R | 1.0 | Tank04 Fill Water Valve Current Status            |        |        |
| Tank04_HaveBasket         | M604    | Bool       | R | 1.0 | Tank04 The Tank has Basket                        |        |        |

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|                        |         |            |   |     |                                                   |        |        |
|------------------------|---------|------------|---|-----|---------------------------------------------------|--------|--------|
| Tank04_RemoteStart     | M331    | Bool       | W | 1.0 | Tank04 Remote Start                               |        |        |
| Tank04_TimePV          | CN104   | Word       | R | 1.0 | Tank04 Time Present Value (s)                     |        |        |
| Tank04_TimeSV          | D4054   | Word       | W | 1.0 | Tank04 Time Set Value (s)                         | 0.0    | 9999.0 |
| Tank04_TimeUp          | CS104   | Bool       | R | 1.0 | Tank04 Time Up                                    |        |        |
| Tank04_WaterFlow       | D1016   | Word       | R | 0.1 | Tank04 Water Flow Meter Value (LPM)               |        |        |
| Tank05_DoorCloseRemote | M305    | Bool       | W | 1.0 | Tank05 Door Remote Close                          |        |        |
| Tank05_DoorClosed      | D1020.5 | Bool       | R | 1.0 | Tank05 Door Actually Closed                       |        |        |
| Tank05_DoorOpenRemote  | M304    | Bool       | W | 1.0 | Tank05 Door Remote Open                           |        |        |
| Tank05_DoorOpened      | D1020.4 | Bool       | R | 1.0 | Tank05 Door Actually Opened                       |        |        |
| Tank05_FillWaterAuto   | L220    | Bool       | W | 1.0 | Tank05 Fill Water Auto                            |        |        |
| Tank05_FillWaterOn     | L120    | Bool       | W | 1.0 | Tank05 Fill Water always On                       |        |        |
| Tank05_FillWaterStatus | D1022.8 | Bool       | R | 1.0 | Tank05 Fill Water Valve Current Status            |        |        |
| Tank05_FilterRemote    | M342    | Bool       | W | 1.0 | Tank05 Filter Remote On                           |        |        |
| Tank05_FilterStatus    | D1022.1 | Bool       | R | 1.0 | Tank05 Filter Pump Status                         |        |        |
| Tank05_HaveBasket      | M605    | Bool       | R | 1.0 | Tank05 The Tank has Basket                        |        |        |
| Tank05_HeaterRemote    | M343    | Bool       | W | 1.0 | Tank05 Heater Remote On                           |        |        |
| Tank05_HeaterStatus    | D1022.2 | Bool       | R | 1.0 | Tank05 Heater Status                              |        |        |
| Tank05_RemoteStart     | M340    | Bool       | W | 1.0 | Tank05 Remote Start                               |        |        |
| Tank05_SwingRemote     | M344    | Bool       | W | 1.0 | Tank05 Swing Remote On                            |        |        |
| Tank05_SwingSV         | D1018   | Word       | W | 1.0 | Tank05 Swing Speed Set Value (CPM)                | 5.0    | 120.0  |
| Tank05_SwingStatus     | D1022.4 | Bool       | R | 1.0 | Tank05 Swing Status                               |        |        |
| Tank05_TempAdj         | D4012   | SignedWord | W | 0.1 | Tank05 Temperature +- Adjust (Degree Celsius)     | -999.0 | 999.9  |
| Tank05_TempPV          | D4002   | Word       | R | 0.1 | Tank05 Temperature Present Value (Degree Celsius) |        |        |
| Tank05_TempSV          | D4022   | Word       | W | 0.1 | Tank05 Temperature Set Value                      | 0.0    | 999.9  |

|                           |         |       |   |     |                                        |     |        |
|---------------------------|---------|-------|---|-----|----------------------------------------|-----|--------|
|                           |         |       |   |     | (Degree Celsius)                       |     |        |
| Tank05_TimePV             | CN105   | Word  | R | 1.0 | Tank05 Time Present Value (s)          |     |        |
| Tank05_TimeSV             | D4055   | Word  | W | 1.0 | Tank05 Time Set Value (s)              | 0.0 | 9999.0 |
| Tank05_TimeUp             | CS105   | Bool  | R | 1.0 | Tank05 Time Up                         |     |        |
| Tank05_WaterFlow          | D1024   | Float | R | 1.0 | Tank05 Water Flow Meter Value (LPM)    |     |        |
| Tank06_AirAgitationAuto   | L222    | Bool  | W | 1.0 | Tank06 Air Agitation Auto              |     |        |
| Tank06_AirAgitationOn     | L122    | Bool  | W | 1.0 | Tank06 Air Agitation Always On         |     |        |
| Tank06_AirAgitationStatus | D1022.8 | Bool  | R | 1.0 | Tank06 Air Agitation Status            |     |        |
| Tank06_FillWaterAuto      | L221    | Bool  | W | 1.0 | Tank06 Fill Water Auto                 |     |        |
| Tank06_FillWaterOn        | L121    | Bool  | W | 1.0 | Tank06 Fill Water always On            |     |        |
| Tank06_FillWaterStatus    | D1022.C | Bool  | R | 1.0 | Tank06 Fill Water Valve Current Status |     |        |
| Tank06_HaveBasket         | M606    | Bool  | R | 1.0 | Tank06 The Tank has Basket             |     |        |
| Tank06_RemoteStart        | M341    | Bool  | W | 1.0 | Tank06 Remote Start                    |     |        |
| Tank06_TimePV             | CN106   | Word  | R | 1.0 | Tank06 Time Present Value (s)          |     |        |
| Tank06_TimeSV             | D4056   | Word  | W | 1.0 | Tank06 Time Set Value (s)              | 0.0 | 9999.0 |
| Tank06_TimeUp             | CS106   | Bool  | R | 1.0 | Tank06 Time Up                         |     |        |
| Tank06_WaterFlow          | D1026   | Word  | R | 0.1 | Tank06 Water Flow Meter Value (LPM)    |     |        |
| Tank07_AirAgitationAuto   | L235    | Bool  | W | 1.0 | Tank07 Air Agitation Auto              |     |        |
| Tank07_AirAgitationOn     | L135    | Bool  | W | 1.0 | Tank07 Air Agitation Always On         |     |        |
| Tank07_AirAgitationStatus | D1030.F | Bool  | R | 1.0 | Tank07 Air Agitation Status            |     |        |
| Tank07_AirFlow            | D1034   | Word  | R | 0.1 | Tank07 Air Flow Meter Value (LPM)      |     |        |
| Tank07_AmperePV           | D1055   | Word  | R | 0.1 | Tank07 Ampere Present Value (A)        |     |        |
| Tank07_AmpereSV           | D1362   | Word  | W | 0.1 | Tank07 Ampere Set Value (A)            | 0.0 | 200.0  |
| Tank07_CoolerAuto         | L231    | Bool  | W | 1.0 | Tank07 Cool Water                      |     |        |

# KAN-PACIFIC

|                        |         |            |   |     |                                                               |        |        |
|------------------------|---------|------------|---|-----|---------------------------------------------------------------|--------|--------|
|                        |         |            |   |     | Circulation<br>Auto                                           |        |        |
| Tank07_CoolerOn        | L131    | Bool       | W | 1.0 | Tank07 Cool<br>Water<br>Circulation<br>Always On              |        |        |
| Tank07_CoolerStatus    | D1030.B | Bool       | R | 1.0 | Tank07 Cool<br>Water<br>Circulation<br>Status                 |        |        |
| Tank07_DoorCloseRemote | M307    | Bool       | W | 1.0 | Tank07 Door<br>Remote Close                                   |        |        |
| Tank07_DoorClosed      | D1028.5 | Bool       | R | 1.0 | Tank07 Door<br>Actually<br>Closed                             |        |        |
| Tank07_DoorOpenRemote  | M306    | Bool       | W | 1.0 | Tank07 Door<br>Remote Open                                    |        |        |
| Tank07_DoorOpened      | D1028.4 | Bool       | R | 1.0 | Tank07 Door<br>Actually<br>Opened                             |        |        |
| Tank07_FillWaterAuto   | L230    | Bool       | W | 1.0 | Tank07 Fill<br>Water Auto                                     |        |        |
| Tank07_FillWaterOn     | L130    | Bool       | W | 1.0 | Tank07 Fill<br>Water always<br>On                             |        |        |
| Tank07_FillWaterStatus | D1030.8 | Bool       | R | 1.0 | Tank07 Fill<br>Water Valve<br>Current<br>Status               |        |        |
| Tank07_FilterRemote    | M353    | Bool       | W | 1.0 | Tank07 Filter<br>Remote On                                    |        |        |
| Tank07_FilterStatus    | D1030.1 | Bool       | R | 1.0 | Tank07 Filter<br>Pump Status                                  |        |        |
| Tank07_HaveBasket      | M607    | Bool       | R | 1.0 | Tank07 The<br>Tank has<br>Basket                              |        |        |
| Tank07_RectRemote      | M354    | Bool       | W | 1.0 | Tank07<br>Rectifier<br>RemoteOn                               |        |        |
| Tank07_RectStatus      | D1030.4 | Bool       | R | 1.0 | Tank07<br>Rectifier<br>Status                                 |        |        |
| Tank07_RemoteStart     | M350    | Bool       | W | 1.0 | Tank07<br>Remote Start                                        |        |        |
| Tank07_TempAdj         | D4013   | SignedWord | W | 0.1 | Tank07<br>Temperature<br>+- Adjust<br>(Degree<br>Celsius)     | -999.0 | 999.9  |
| Tank07_TempPV          | D4003   | Word       | R | 0.1 | Tank07<br>Temperature<br>Present Value<br>(Degree<br>Celsius) |        |        |
| Tank07_TempSV          | D4023   | Word       | W | 0.1 | Tank07<br>Temperature<br>Set Value<br>(Degree<br>Celsius)     | 0.0    | 999.9  |
| Tank07_TimePV          | CN107   | Word       | R | 1.0 | Tank07 Time<br>Present Value<br>(s)                           |        |        |
| Tank07_TimeSV          | D4057   | Word       | W | 1.0 | Tank07 Time<br>Set Value (s)                                  | 0.0    | 9999.0 |
| Tank07_TimeUp          | CS107   | Bool       | R | 1.0 | Tank07 Time<br>Up                                             |        |        |

|                           |         |       |   |     |                                            |     |        |
|---------------------------|---------|-------|---|-----|--------------------------------------------|-----|--------|
| Tank07_VoltagePV          | D1054   | Word  | R | 0.1 | Tank07 Rectifier Voltage Present Value (V) |     |        |
| Tank07_WaterFlow          | D1032   | Float | R | 1.0 | Tank07 Water Flow Meter Value (LPM)        |     |        |
| Tank08_AirAgitationAuto   | L232    | Bool  | W | 1.0 | Tank08 Air Agitation Auto                  |     |        |
| Tank08_AirAgitationOn     | L132    | Bool  | W | 1.0 | Tank08 Air Agitation Always On             |     |        |
| Tank08_AirAgitationStatus | D1030.C | Bool  | R | 1.0 | Tank08 Air Agitation Status                |     |        |
| Tank08_HaveBasket         | M608    | Bool  | R | 1.0 | Tank08 The Tank has Basket                 |     |        |
| Tank08_RemoteStart        | M351    | Bool  | W | 1.0 | Tank08 Remote Start                        |     |        |
| Tank08_TimePV             | CN108   | Word  | R | 1.0 | Tank08 Time Present Value (s)              |     |        |
| Tank08_TimeSV             | D4058   | Word  | W | 1.0 | Tank08 Time Set Value (s)                  | 0.0 | 9999.0 |
| Tank08_TimeUp             | CS108   | Bool  | R | 1.0 | Tank08 Time Up                             |     |        |
| Tank08_WaterFlow          | D1036   | Word  | R | 0.1 | Tank08 Water Flow Meter Value (LPM)        |     |        |
| Tank09_AirAgitationAuto   | L233    | Bool  | W | 1.0 | Tank09 Air Agitation Auto                  |     |        |
| Tank09_AirAgitationOn     | L133    | Bool  | W | 1.0 | Tank09 Air Agitation Always On             |     |        |
| Tank09_AirAgitationStatus | D1030.D | Bool  | R | 1.0 | Tank09 Air Agitation Status                |     |        |
| Tank09_FillWaterAuto      | L234    | Bool  | W | 1.0 | Tank09 Fill Water Auto                     |     |        |
| Tank09_FillWaterOn        | L134    | Bool  | W | 1.0 | Tank09 Fill Water always On                |     |        |
| Tank09_FillWaterStatus    | D1030.E | Bool  | R | 1.0 | Tank09 Fill Water Valve Current Status     |     |        |
| Tank09_HaveBasket         | M609    | Bool  | R | 1.0 | Tank09 The Tank has Basket                 |     |        |
| Tank09_RemoteStart        | M352    | Bool  | W | 1.0 | Tank09 Remote Start                        |     |        |
| Tank09_TimePV             | CN109   | Word  | R | 1.0 | Tank09 Time Present Value (s)              |     |        |
| Tank09_TimeSV             | D4059   | Word  | W | 1.0 | Tank09 Time Set Value (s)                  | 0.0 | 9999.0 |
| Tank09_TimeUp             | CS109   | Bool  | R | 1.0 | Tank09 Time Up                             |     |        |
| Tank09_WaterFlow          | D1036   | Word  | R | 0.1 | Tank09 Water Flow Meter Value (LPM)        |     |        |

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|                        |         |            |   |      |                                                   |        |        |
|------------------------|---------|------------|---|------|---------------------------------------------------|--------|--------|
| Tank10_DoorCloseRemote | M309    | Bool       | W | 1.0  | Tank10 Door Remote Close                          |        |        |
| Tank10_DoorClosed      | D1102.5 | Bool       | R | 1.0  | Tank10 Door Actually Closed                       |        |        |
| Tank10_DoorOpenRemote  | M308    | Bool       | W | 1.0  | Tank10 Door Remote Open                           |        |        |
| Tank10_DoorOpened      | D1102.4 | Bool       | R | 1.0  | Tank10 Door Actually Opened                       |        |        |
| Tank10_FillWaterAuto   | L240    | Bool       | W | 1.0  | Tank10 Fill Water Auto                            |        |        |
| Tank10_FillWaterOn     | L140    | Bool       | W | 1.0  | Tank10 Fill Water always On                       |        |        |
| Tank10_FillWaterStatus | D1104.8 | Bool       | R | 1.0  | Tank10 Fill Water Valve Current Status            |        |        |
| Tank10_FilterRemote    | M361    | Bool       | W | 1.0  | Tank10 Filter Remote On                           |        |        |
| Tank10_FilterStatus    | D1104.1 | Bool       | R | 1.0  | Tank10 Filter Pump Status                         |        |        |
| Tank10_HaveBasket      | M610    | Bool       | R | 1.0  | Tank10 The Tank has Basket                        |        |        |
| Tank10_HeaterRemote    | M362    | Bool       | W | 1.0  | Tank10 Heater Remote On                           |        |        |
| Tank10_HeaterStatus    | D1104.2 | Bool       | R | 1.0  | Tank10 Heater Status                              |        |        |
| Tank10_PH              | D1108   | Word       | R | 0.01 | Tank10 PH Meter Value                             |        |        |
| Tank10_RemoteStart     | M360    | Bool       | W | 1.0  | Tank10 Remote Start                               |        |        |
| Tank10_SwingRemote     | M363    | Bool       | W | 1.0  | Tank10 Swing Remote On                            |        |        |
| Tank10_SwingSV         | D1100   | Word       | W | 1.0  | Tank10 Swing Speed Set Value (CPM)                | 5.0    | 120.0  |
| Tank10_SwingStatus     | D1104.3 | Bool       | R | 1.0  | Tank10 Swing Status                               |        |        |
| Tank10_TempAdj         | D4014   | SignedWord | W | 0.1  | Tank10 Temperature +- Adjust (Degree Celsius)     | -999.0 | 999.9  |
| Tank10_TempPV          | D4004   | Word       | R | 0.1  | Tank10 Temperature Present Value (Degree Celsius) |        |        |
| Tank10_TempSV          | D4024   | Word       | W | 0.1  | Tank10 Temperature Set Value (Degree Celsius)     | 0.0    | 999.9  |
| Tank10_TimePV          | CN110   | Word       | R | 1.0  | Tank10 Time Present Value (s)                     |        |        |
| Tank10_TimeSV          | D4060   | Word       | W | 1.0  | Tank10 Time Set Value (s)                         | 0.0    | 9999.0 |
| Tank10_TimeUp          | CS110   | Bool       | R | 1.0  | Tank10 Time Up                                    |        |        |

# KAN-PACIFIC

|                           |         |       |   |     |                                        |     |        |
|---------------------------|---------|-------|---|-----|----------------------------------------|-----|--------|
| Tank10_WaterFlow          | D1106   | Float | R | 1.0 | Tank10 Water Flow Meter Value (LPM)    |     |        |
| Tank11_FillWaterAuto      | L250    | Bool  | W | 1.0 | Tank11 Fill Water Auto                 |     |        |
| Tank11_FillWaterOn        | L150    | Bool  | W | 1.0 | Tank11 Fill Water always On            |     |        |
| Tank11_FillWaterStatus    | D1112.4 | Bool  | R | 1.0 | Tank11 Fill Water Valve Current Status |     |        |
| Tank11_HaveBasket         | M611    | Bool  | R | 1.0 | Tank11 The Tank has Basket             |     |        |
| Tank11_RemoteStart        | M370    | Bool  | W | 1.0 | Tank11 Remote Start                    |     |        |
| Tank11_SprayRemote        | M371    | Bool  | W | 1.0 | Tank11 Spray Remote On                 |     |        |
| Tank11_SprayStatus        | D1112.1 | Bool  | R | 1.0 | Tank11 Spray Status                    |     |        |
| Tank11_TimePV             | CN111   | Word  | R | 1.0 | Tank11 Time Present Value (s)          |     |        |
| Tank11_TimeSV             | D4061   | Word  | W | 1.0 | Tank11 Time Set Value (s)              | 0.0 | 9999.0 |
| Tank11_TimeUp             | CS111   | Bool  | R | 1.0 | Tank11 Time Up                         |     |        |
| Tank11_WaterFlow          | D1114   | Word  | R | 0.1 | Tank11 Water Flow Meter Value (LPM)    |     |        |
| Tank12_AirAgitationAuto   | L261    | Bool  | W | 1.0 | Tank12 Air Agitation Auto              |     |        |
| Tank12_AirAgitationOn     | L161    | Bool  | W | 1.0 | Tank12 Air Agitation Always On         |     |        |
| Tank12_AirAgitationStatus | D1118.4 | Bool  | R | 1.0 | Tank12 Air Agitation Status            |     |        |
| Tank12_DoorCloseRemote    | M311    | Bool  | W | 1.0 | Tank12 Door Remote Close               |     |        |
| Tank12_DoorClosed         | D1116.5 | Bool  | R | 1.0 | Tank12 Door Actually Closed            |     |        |
| Tank12_DoorOpenRemote     | M310    | Bool  | W | 1.0 | Tank12 Door Remote Open                |     |        |
| Tank12_DoorOpened         | D1116.4 | Bool  | R | 1.0 | Tank12 Door Actually Opened            |     |        |
| Tank12_FillWaterAuto      | L262    | Bool  | W | 1.0 | Tank12 Fill Water Auto                 |     |        |
| Tank12_FillWaterOn        | L162    | Bool  | W | 1.0 | Tank12 Fill Water always On            |     |        |
| Tank12_FillWaterStatus    | D1118.5 | Bool  | R | 1.0 | Tank12 Fill Water Valve Current Status |     |        |
| Tank12_HaveBasket         | M612    | Bool  | R | 1.0 | Tank12 The Tank has Basket             |     |        |
| Tank12_RemoteStart        | M380    | Bool  | W | 1.0 | Tank12 Remote Start                    |     |        |

|                        |         |            |   |     |                                                   |        |        |
|------------------------|---------|------------|---|-----|---------------------------------------------------|--------|--------|
| Tank12_TimePV          | CN112   | Word       | R | 1.0 | Tank12 Time Present Value (s)                     |        |        |
| Tank12_TimeSV          | D4062   | Word       | W | 1.0 | Tank12 Time Set Value (s)                         | 0.0    | 9999.0 |
| Tank12_TimeUp          | CS112   | Bool       | R | 1.0 | Tank12 Time Up                                    |        |        |
| Tank12_WaterFlow       | D1120   | Word       | R | 0.1 | Tank12 Water Flow Meter Value (LPM)               |        |        |
| Tank13_Conductivity    | D1122   | Float      | R | 1.0 | Conductivity Meter Reading (us/cm)                |        |        |
| Tank13_FillWaterAuto   | L260    | Bool       | W | 1.0 | Tank13 Fill Water Auto                            |        |        |
| Tank13_FillWaterOn     | L160    | Bool       | W | 1.0 | Tank13 Fill Water always On                       |        |        |
| Tank13_FillWaterStatus | D1118.3 | Bool       | R | 1.0 | Tank13 Fill Water Valve Current Status            |        |        |
| Tank13_HaveBasket      | M613    | Bool       | R | 1.0 | Tank13 The Tank has Basket                        |        |        |
| Tank13_HeaterRemote    | M382    | Bool       | W | 1.0 | Tank13 Heater Remote On                           |        |        |
| Tank13_HeaterStatus    | D1118.2 | Bool       | R | 1.0 | Tank13 Heater Status                              |        |        |
| Tank13_RemoteStart     | M381    | Bool       | W | 1.0 | Tank13 Remote Start                               |        |        |
| Tank13_TempAdj         | D4015   | SignedWord | W | 0.1 | Tank13 Temperature +- Adjust (Degree Celsius)     | -999.0 | 999.9  |
| Tank13_TempPV          | D4005   | Word       | R | 0.1 | Tank13 Temperature Present Value (Degree Celsius) |        |        |
| Tank13_TempSV          | D4025   | Word       | W | 0.1 | Tank13 Temperature Set Value (Degree Celsius)     | 0.0    | 999.9  |
| Tank13_TimePV          | CN113   | Word       | R | 1.0 | Tank13 Time Present Value (s)                     |        |        |
| Tank13_TimeSV          | D4063   | Word       | W | 1.0 | Tank13 Time Set Value (s)                         | 0.0    | 9999.0 |
| Tank13_TimeUp          | CS113   | Bool       | R | 1.0 | Tank13 Time Up                                    |        |        |
| Tank14_DoorCloseRemote | M313    | Bool       | W | 1.0 | Tank14 Door Remote Close                          |        |        |
| Tank14_DoorClosed      | D1126.5 | Bool       | R | 1.0 | Tank14 Door Actually Closed                       |        |        |
| Tank14_DoorOpenRemote  | M312    | Bool       | W | 1.0 | Tank14 Door Remote Open                           |        |        |
| Tank14_DoorOpened      | D1126.4 | Bool       | R | 1.0 | Tank14 Door Actually Opened                       |        |        |

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|---------------------------|---------|------------|---|------|---------------------------------------------------|--------|--------|
| Tank14_FillWaterAuto      | L270    | Bool       | W | 1.0  | Tank14 Fill Water Auto                            |        |        |
| Tank14_FillWaterOn        | L170    | Bool       | W | 1.0  | Tank14 Fill Water always On                       |        |        |
| Tank14_FillWaterStatus    | D1128.8 | Bool       | R | 1.0  | Tank14 Fill Water Valve Current Status            |        |        |
| Tank14_FilterRemote       | M391    | Bool       | W | 1.0  | Tank14 Filter Remote On                           |        |        |
| Tank14_FilterStatus       | D1128.1 | Bool       | R | 1.0  | Tank14 Filter Pump Status                         |        |        |
| Tank14_HaveBasket         | M614    | Bool       | R | 1.0  | Tank14 The Tank has Basket                        |        |        |
| Tank14_HeaterRemote       | M392    | Bool       | W | 1.0  | Tank14 Heater Remote On                           |        |        |
| Tank14_HeaterStatus       | D1128.2 | Bool       | R | 1.0  | Tank14 Heater Status                              |        |        |
| Tank14_PH                 | D1132   | Word       | R | 0.01 | Tank14 PH Meter Value                             |        |        |
| Tank14_RemoteStart        | M390    | Bool       | W | 1.0  | Tank14 Remote Start                               |        |        |
| Tank14_SwingRemote        | M393    | Bool       | W | 1.0  | Tank14 Swing Remote On                            |        |        |
| Tank14_SwingSV            | D1124   | Word       | W | 1.0  | Tank14 Swing Speed Set Value (CPM)                | 5.0    | 120.0  |
| Tank14_SwingStatus        | D1128.3 | Bool       | R | 1.0  | Tank14 Swing Status                               |        |        |
| Tank14_TempAdj            | D4016   | SignedWord | W | 0.1  | Tank14 Temperature +- Adjust (Degree Celsius)     | -999.0 | 999.9  |
| Tank14_TempPV             | D4006   | Word       | R | 0.1  | Tank14 Temperature Present Value (Degree Celsius) |        |        |
| Tank14_TempSV             | D4026   | Word       | W | 0.1  | Tank14 Temperature Set Value (Degree Celsius)     | 0.0    | 999.9  |
| Tank14_TimePV             | CN114   | Word       | R | 1.0  | Tank14 Time Present Value (s)                     |        |        |
| Tank14_TimeSV             | D4064   | Word       | W | 1.0  | Tank14 Time Set Value (s)                         | 0.0    | 9999.0 |
| Tank14_TimeUp             | CS114   | Bool       | R | 1.0  | Tank14 Time Up                                    |        |        |
| Tank14_WaterFlow          | D1130   | Float      | R | 1.0  | Tank14 Water Flow Meter Value (LPM)               |        |        |
| Tank15_AirAgitationAuto   | L280    | Bool       | W | 1.0  | Tank15 Air Agitation Auto                         |        |        |
| Tank15_AirAgitationOn     | L180    | Bool       | W | 1.0  | Tank15 Air Agitation Always On                    |        |        |
| Tank15_AirAgitationStatus | D1136.4 | Bool       | R | 1.0  | Tank15 Air Agitation Status                       |        |        |

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|                           |         |            |   |     |                                                |        |        |
|---------------------------|---------|------------|---|-----|------------------------------------------------|--------|--------|
| Tank15_FillWaterAuto      | L281    | Bool       | W | 1.0 | Tank15 Fill Water Auto                         |        |        |
| Tank15_FillWaterOn        | L181    | Bool       | W | 1.0 | Tank15 Fill Water always On                    |        |        |
| Tank15_FillWaterStatus    | D1136.5 | Bool       | R | 1.0 | Tank15 Fill Water Valve Current Status         |        |        |
| Tank15_HaveBasket         | M615    | Bool       | R | 1.0 | Tank15 The Tank has Basket                     |        |        |
| Tank15_RemoteStart        | M400    | Bool       | W | 1.0 | Tank15 Remote Start                            |        |        |
| Tank15_TimePV             | CN115   | Word       | R | 1.0 | Tank15 Time Present Value (s)                  |        |        |
| Tank15_TimeSV             | D4065   | Word       | W | 1.0 | Tank15 Time Set Value (s)                      | 0.0    | 9999.0 |
| Tank15_TimeUp             | CS115   | Bool       | R | 1.0 | Tank15 Time Up                                 |        |        |
| Tank15_WaterFlow          | D1138   | Word       | R | 0.1 | Tank15 Water Flow Meter Value (LPM)            |        |        |
| Tank16_AirAgitationAuto   | L282    | Bool       | W | 1.0 | Tank16 Air Agitation Auto                      |        |        |
| Tank16_AirAgitationOn     | L182    | Bool       | W | 1.0 | Tank16 Air Agitation Always On                 |        |        |
| Tank16_AirAgitationStatus | D1136.6 | Bool       | R | 1.0 | Tank16 Air Agitation Status                    |        |        |
| Tank16_Conductivity       | D1142   | Float      | R | 1.0 | Conductivity Meter Reading (MOhmCM)            |        |        |
| Tank16_FillWaterAuto      | L283    | Bool       | W | 1.0 | Tank16 Fill Water Auto                         |        |        |
| Tank16_FillWaterOn        | L183    | Bool       | W | 1.0 | Tank16 Fill Water always On                    |        |        |
| Tank16_FillWaterStatus    | D1136.7 | Bool       | R | 1.0 | Tank16 Fill Water Valve Current Status         |        |        |
| Tank16_FilterRemote       | M402    | Bool       | W | 1.0 | Tank16 Filter Remote On                        |        |        |
| Tank16_FilterStatus       | D1136.2 | Bool       | R | 1.0 | Tank16 Filter Pump Status                      |        |        |
| Tank16_HaveBasket         | M616    | Bool       | R | 1.0 | Tank16 The Tank has Basket                     |        |        |
| Tank16_HeaterRemote       | M403    | Bool       | W | 1.0 | Tank16 Heater Remote On                        |        |        |
| Tank16_HeaterStatus       | D1136.3 | Bool       | R | 1.0 | Tank16 Heater Status                           |        |        |
| Tank16_RemoteStart        | M401    | Bool       | W | 1.0 | Tank16 Remote Start                            |        |        |
| Tank16_TempAdj            | D4017   | SignedWord | W | 0.1 | Tank16 Temperature +/- Adjust (Degree Celsius) | -999.0 | 999.9  |

|                  |       |      |   |     |                                                   |     |        |
|------------------|-------|------|---|-----|---------------------------------------------------|-----|--------|
| Tank16_TempPV    | D4007 | Word | R | 0.1 | Tank16 Temperature Present Value (Degree Celsius) |     |        |
| Tank16_TempSV    | D4027 | Word | W | 0.1 | Tank16 Temperature Set Value (Degree Celsius)     | 0.0 | 999.9  |
| Tank16_TimePV    | CN116 | Word | R | 1.0 | Tank16 Time Present Value (s)                     |     |        |
| Tank16_TimeSV    | D4066 | Word | W | 1.0 | Tank16 Time Set Value (s)                         | 0.0 | 9999.0 |
| Tank16_TimeUp    | CS116 | Bool | R | 1.0 | Tank16 Time Up                                    |     |        |
| Tank16_WaterFlow | D1140 | Word | R | 0.1 | Tank16 Water Flow Meter Value (LPM)               |     |        |

## 9. Client Examples

### 9.1 curl

```
curl -H "X-API-Key: YOUR_KEY" http://127.0.0.1:8000/api/v1/tags/list
```

```
curl -H "X-API-Key: YOUR_KEY" http://127.0.0.1:8000/api/v1/tags/read
```

```
curl -H "X-API-Key: YOUR_KEY" -H "Content-Type: application/json" \
-d '{"tag_name":"Tank01_TempSV","value":25.0}' \
http://127.0.0.1:8000/api/v1/tags/write
```

### 9.2 Python

```
import json
import urllib.request

key = open("dev_api_key.txt").read().strip()
req = urllib.request.Request(
    "http://127.0.0.1:8000/api/v1/tags/read",
    headers={"X-API-Key": key},
)
with urllib.request.urlopen(req) as resp:
    print(json.loads(resp.read()))
```

## 10. Error Codes

| Status | Typical cause                                                                |
|--------|------------------------------------------------------------------------------|
| 400    | Invalid value, out of Low/High limit, unsupported type, duplicate batch tags |
| 401    | Missing/invalid X-API-Key                                                    |
| 403    | Tag is read-only (ReadWrite=R)                                               |
| 404    | Tag not found, or no poll records yet                                        |
| 502    | PLC write/communication failure                                              |

|     |                                                         |
|-----|---------------------------------------------------------|
| 503 | API key missing, database missing, PLC busy/unavailable |
|-----|---------------------------------------------------------|

## 11. Troubleshooting

| Problem                    | What to do                                       |
|----------------------------|--------------------------------------------------|
| 503 API key not configured | Run python generate_api_key.py                   |
| 503 Database not found     | Run plc_polling.py or set PLC_DB_PATH            |
| 503 PLC is busy            | Retry after the current poll cycle finishes      |
| 401 Invalid API key        | Match X-API-Key to dev_api_key.txt / PLC_API_KEY |
| 404 No polling records     | Start the poller and wait for one cycle          |
| 400 below/above limit      | Check /tags/list low_limit and high_limit        |
| Port already in use        | Set PLC_API_PORT to another port                 |

## 12. Related Files

| File                    | Purpose                                            |
|-------------------------|----------------------------------------------------|
| plc_api.py              | FastAPI application                                |
| plc_polling.py          | Polls PLC and fills the database                   |
| plc_config.py           | PLC IP/port and TagsMap loading                    |
| plc_connection_guard.py | Shared PLC lock between poller and API             |
| TagsMap.xlsx            | Tag definitions, multipliers, limits, descriptions |
| generate_api_key.py     | Create development API key                         |
| dev_api_key.txt         | Local API key (do not commit)                      |
| plc_polling_data.db     | SQLite store for poll results                      |

## 13. Security Notes

- The shared API key is intended for development/local use.
- Do not expose 0.0.0.0 on untrusted networks without HTTPS and stronger auth.
- Rotate the key with generate\_api\_key.py if it is leaked.
- For production, add TLS, stronger authentication, and rate limiting.

End of PLC API User Manual